2. Fields

Observe three field states

Build a force map from repeated compass tests

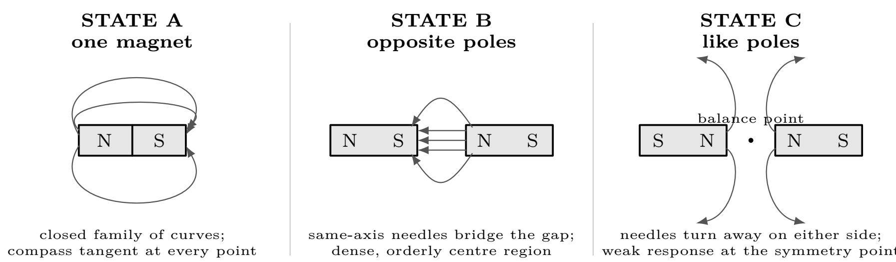
Lesson 2 · two sessions of 45 minutes · learner maps; adult manages filings

What you need

- Two labelled bar magnets
- One small compass; a second compass for the proof test
- Graph paper, ruler, pencil, and protractor
- Clear acrylic or stiff card
- Iron filings sealed in a shaker
- Tape to mark fixed magnet positions

Predict first

Draw the compass direction at the centre of each arrangement before testing: one magnet; opposite poles facing; like poles facing. For each prediction, write one observation that would make you change your mind.



Read the states before drawing lines

A. Where is the compass turn largest? **B.** Does the centre needle point across or along the gap? **C.** At the balance point, does “small response” mean “no field anywhere”? Answer from nearby compass readings, not from filings alone.

First pass: reveal, then reset

Tape magnet outlines to the bench so each state can be reproduced. The adult shakes a sparse layer of filings onto the card; the learner taps the card, sketches the pattern, and labels the magnet positions. Return filings to their container before changing state. Keep filings away from eyes, electronics, and the compass pivot. Filings show alignment and crowding, but they do not supply arrow direction; the compass does.

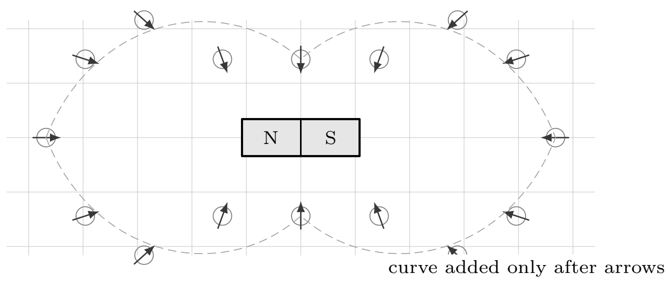
2. Fields

Walk one compass; keep every trial

Build a force map from repeated compass tests

Build a point-by-point map

1. Tape one bar magnet on a graph-paper grid. Trace its outline and mark which end is north. Do not move it until the table is complete.
2. Place the compass centre on a grid intersection, wait for the needle to settle, and trace the north-pointing direction as an arrow.
3. Lift the compass completely, replace it at the same point, and repeat. Record both bearings. A repeat is evidence about reliability, not busywork.
4. Continue around the magnet. Use the same compass, reading position, settling time, and grid spacing at every point.
5. Only after all arrows exist, add faint curves tangent to them. Never bend a curve merely to make the picture look symmetrical.



Questions while the map is still raw

1. Which two points should match by symmetry? Do their repeated bearings agree within your reading precision?
2. Near a pole, does the needle settle faster, slower, or no differently? Record the observation; do not turn settling speed directly into field strength.
3. If two nearby arrows disagree sharply, should a smooth curve pass through both? What repeat would decide which arrow to trust?

Repeated bearing record

Measure bearings clockwise from the top edge of the paper. Use identical grid points for all repeats.

Point	x (cm)	y (cm)	bearing 1	bearing 2	difference	observation
A						
B						
C						
D						
E						
F						

Stop and judge the repeats

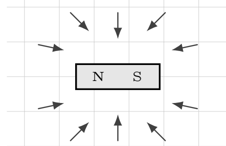
Circle the largest bearing difference. If it exceeds 10°, make a third reading before drawing any guide curve. Check first for a shifted magnet, nearby steel, a tilted compass, or a grid point read from the wrong side.

2. Fields

Turn observations into a model

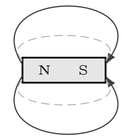
Build a force map from repeated compass tests

MEASURED STATE



dots + needle bearings
remain the primary evidence

INFERRED MAP



smooth curves summarize
direction between samples

Worked inference

At the grid point $(x, y) = (0, 3\,\mathrm{cm})$, suppose two north-needle bearings are 88° and 92°. Their mean is

$$(88^\circ + 92^\circ)/2 = 90^\circ,$$

and their spread is 4°. The defensible plotted result is an arrow at about 90°, with the repeat spread noted. It supports a horizontal field direction at that point. It does *not* measure field magnitude: a freely rotating compass reports direction, while its turn angle relative to the paper depends on the combined magnet and Earth's fields.

A second inference: comparison, not a law

Suppose a compass initially points 0° with the magnet absent. On the same axis it turns to 58° at $2\,\mathrm{cm}$, 31° at $4\,\mathrm{cm}$, and 17° at $6\,\mathrm{cm}$. The response decreases with distance. That trend is supported. "The field halves whenever distance doubles" is not: 58° to 31° is only one comparison, and compass angle is not directly proportional to magnetic-field strength.

Change one state, preserve the test

Repeat four fixed grid points for opposite poles facing and like poles facing. Use the same gap and compass.

Point	opposite 1	opposite 2	like 1	like 2	what changed?
centre					
above centre					
left of gap					
right of gap					

Intermediate claim check

For each state, write: "At point ___, the needle pointed ___; therefore the field direction there was ___." Then add: "This does not yet tell me ___." A strong claim says both what the observation establishes and what it leaves open.

Electric-field analogy

The map-making rule transfers: an electric-field arrow gives the force direction on a positive test charge at one location. The probes differ. A compass samples a magnetic field; it does not sample electric field. Electric-field lines begin on positive charge and end on negative charge, whereas magnetic-field lines form closed loops. In both cases, drawn lines are a model assembled from local evidence, not visible threads in space.

2. Fields

Diagnose the map; then make it predict

Build a force map from repeated compass tests

Diagnostics

Symptom	Likely cause	Correction
All needles point alike	magnet too far away; steel object dominates	bring the fixed grid closer; clear the bench
Repeat differs greatly	compass not centred; magnet shifted	mark centres; retape outline; take a third reading
Needle sticks	tilted housing or filings at pivot	level and clean compass; replace if needed
Curves cross	curves invented between sparse samples	erase curves; collect another local arrow
Map looks asymmetric	real placement error or biased reading	swap observer side; remeasure mirrored points
Weak centre in like-pole state	symmetry cancellation near that point	test immediately above, below, left, and right

Final proof: predict a hidden reading

1. Choose a blank grid point between four measured points. Without placing a compass there, draw the direction your map predicts. Write a tolerance, such as "within 10°."
2. Give the sheet and a second compass to another person. Hide your prediction with a card while they centre the compass and record two bearings.
3. Reveal the prediction. Compute the smallest angular difference between prediction and each reading. A prediction of 350° and reading of 5° differ by 15°, not 345°.
4. Repeat at a point where the map bends sharply. State whether each result falls within tolerance. If not, revise the local curve or collect more points; do not revise the recorded bearings.

Proof record

Point + prediction	__________	Tolerance	__________
Hidden readings	__________	Difference	__________
Pass / revise	__________	Reason	__________

What the proof establishes

A successful hidden prediction shows that the field map compresses earlier measurements well enough to predict a new local direction. It does not prove that the drawn curves are physical strands, that the map is exact between all points, or that magnetic and electric fields are the same thing.

The link to the coil

The primary coil creates a magnetic field; the secondary responds to the field at its own location. Later lessons change that field with time. This map has already established the essential discipline: fix the source, sample defined locations, preserve repeats, infer only what the probe can show, then demand a new prediction.

> **This lesson uses no electrical power source.** Keep loose filings away from eyes, electronics, and the compass pivot. The habits that matter here are experimental: fix the source, change one state at a time, preserve every reading, and test the finished model against a new observation.